

A dark gray L-shaped frame composed of two thick lines. One line runs vertically down the left side, and the other runs horizontally across the top, meeting at a right angle in the upper-left corner. Another L-shaped line runs vertically down the right side and horizontally across the bottom, meeting at a right angle in the lower-right corner.

ENGINEERING IN THE GLOBAL CONTEXT

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Case



Introduction

- What are the most common ethical issues faced by engineers globally?

The Movement Toward Globalized Engineering Educational Standards

- Washington Accord – 1989
- Universal criteria for engineering education
- Mutual recognition of qualifications
- Signatories:
 - United Kingdom, Ireland, the United States, Canada, Australia, New Zealand, China, Chinese Taipei, Hong Kong China, India, Japan, Korea, Malaysia, Russia, Singapore, South Africa, Sri Lanka, Turkey, Bangladesh, Costa Rica, Indonesia, Mexico, Pakistan, Peru, Philippines

International Professionalism and Ethics

- Consensus on whether engineers are professionals?
- What professionalism might mean?
- Do engineering societies call their members professional?
- Do engineers worldwide have the concept of “professional”?

Boundary-crossing Problems

- The problems encountered when one crosses national and cultural borders
- Absolutist solution or imperialist solution - to hold to home-country values and ways of doing things
- Relativist solution – “When in Rome, do as the Romans do.”
- How standards should be applied?
- Moral laxism – sometimes moral principles appear so far removed from the moral issue at hand that they cannot be applied with any precision, so that almost any action is permissible
- Moral rigorism - moral principles, whether they are those of the home country or host country, must be strictly applied in every situation

Ethical Resources for Globalized Engineering

- Creative middle ways
- The golden rule
- Dignity: universal human rights
- Development: promoting basic human well-being
- The resources of virtue ethics
- Codes of engineering societies

Economic Underdevelopment: The Problem of Exploitation

- Conditions for Exploitation:
- An imbalance of (usually economic) power between the dominant and subordinate or exploited party.
- The subordinate party needs the resources by the dominant party to protect his or her vital interests.
- For the subordinate party, the exploitative relationship is the only source of such resources.
- The dominant party in the relationship exercises discretionary control over the needed resources.
- The resources of the subordinate party (natural resources, labor, etc.) are used without adequate compensation.

Paying for Special Treatment: The Problem of Bribery

- A bribe is a payment of money (or something of value) to another person in exchange for his giving special consideration that is incompatible with the duties of his office, position, or role.
- Bribes presuppose an agreement that the bribe must be in exchange for a certain kind of conduct.
- If this agreement is not present, then it is difficult to distinguish bribes from gifts or rewards.

Paying for Deserved Services: The Problem of Extortion and Grease Payments

- Extortion is the act of threatening someone with harm (that the extorter is not entitled to inflict) to obtain benefits to which the extorter has no prior right.
- Grease payments are typically offered to facilitate routine bureaucratic decisions, such as hastening the passage of goods through customs or getting faster processing of permits.

The Extended Family Unit: The Problem of Nepotism

- In many areas of the world, the primary unit of society is not the individual, as it is in the modern West, but some larger group.
- The larger group may be an extended family, which includes brothers and sisters and their families, aunts, uncles, cousins, and so forth.
- If a member of the group has the good fortune to get a well-paying job, he or she has an obligation to find jobs for his or her relatives.
- However, this may produce problems.

The Absence of Technical-scientific Sophistication: The Problem of Paternalism

- Paternalism occurs when individuals (including engineers), governments, or corporations override the ability of others to decide what they should (or should not) do in the interests of those others.

BOX 8.7 Two Types of Paternalism

- **Weak paternalism.** The paternalist overrides the decision-making powers of the recipient when there is reason to believe the recipient is not able to exercise her moral agency effectively.
- **Strong paternalism.** The paternalist overrides the decision-making powers of the recipient, even when there is no reason to believe the recipient is not exercising his or her moral agency effectively.